

Analysis

2024-10-10

```
library(ggplot2)
```

Union station

```
data <- read.csv('./CHI.csv')
```

```
data$majority <- relevel(factor(data$majority), ref='white')
```

```
#head(data)
```

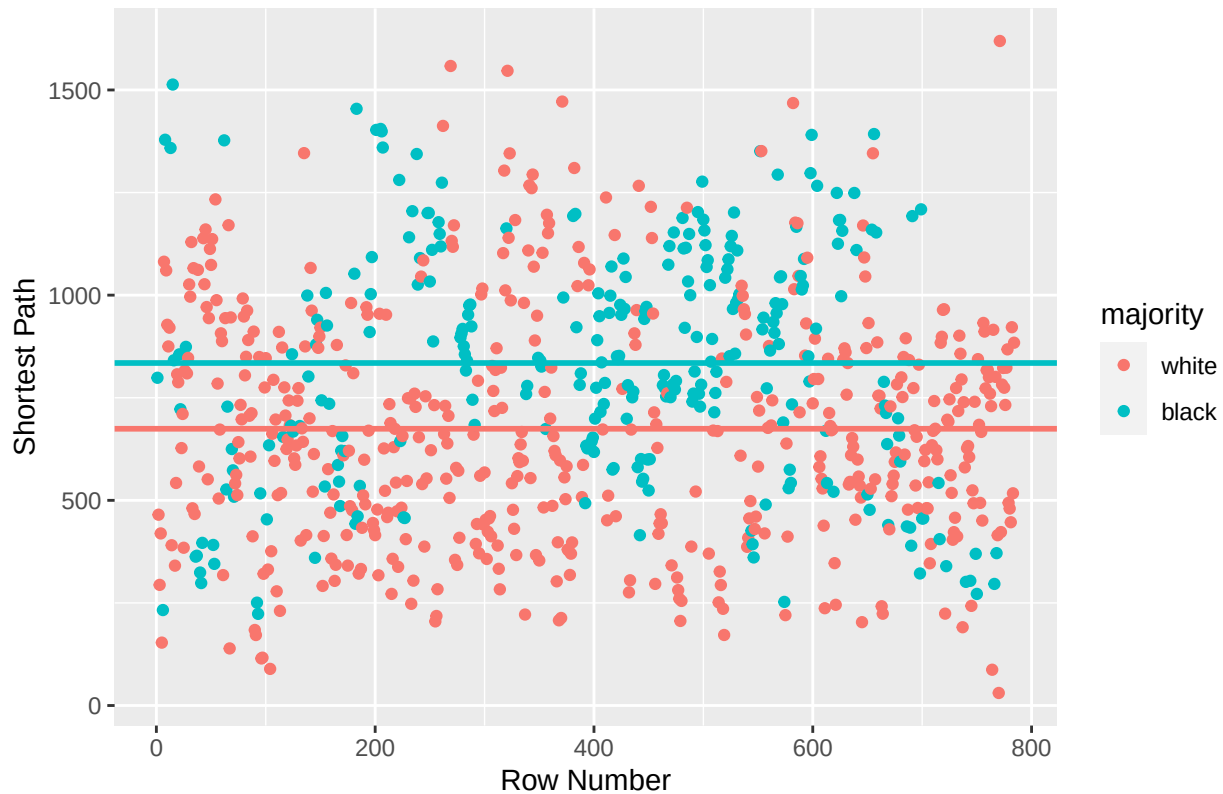
```
black_mean_ttr <- mean(data[data$majority == 'black', ]$shortest_path)
```

```
white_mean_ttr <- mean(data[data$majority == 'white', ]$shortest_path)
```

Mean time to reach by demographic group

```
ggplot(data, aes(x = seq_along(shortest_path), y = shortest_path, color = majority)) +  
  geom_point() +  
  geom_hline(yintercept=black_mean_ttr, linewidth=1, color="#00bfc4") +  
  geom_hline(yintercept=white_mean_ttr, linewidth=1, color="#F8766D") +  
  labs(x = "Row Number", y = "Shortest Path",  
       title = "Time to reach destination by demographic group")
```

Time to reach destination by demographic group



```
results <- t.test(data[data$majority == 'black', ]$shortest_path, data[data$majority == 'white', ]$shortest_path)
results
```

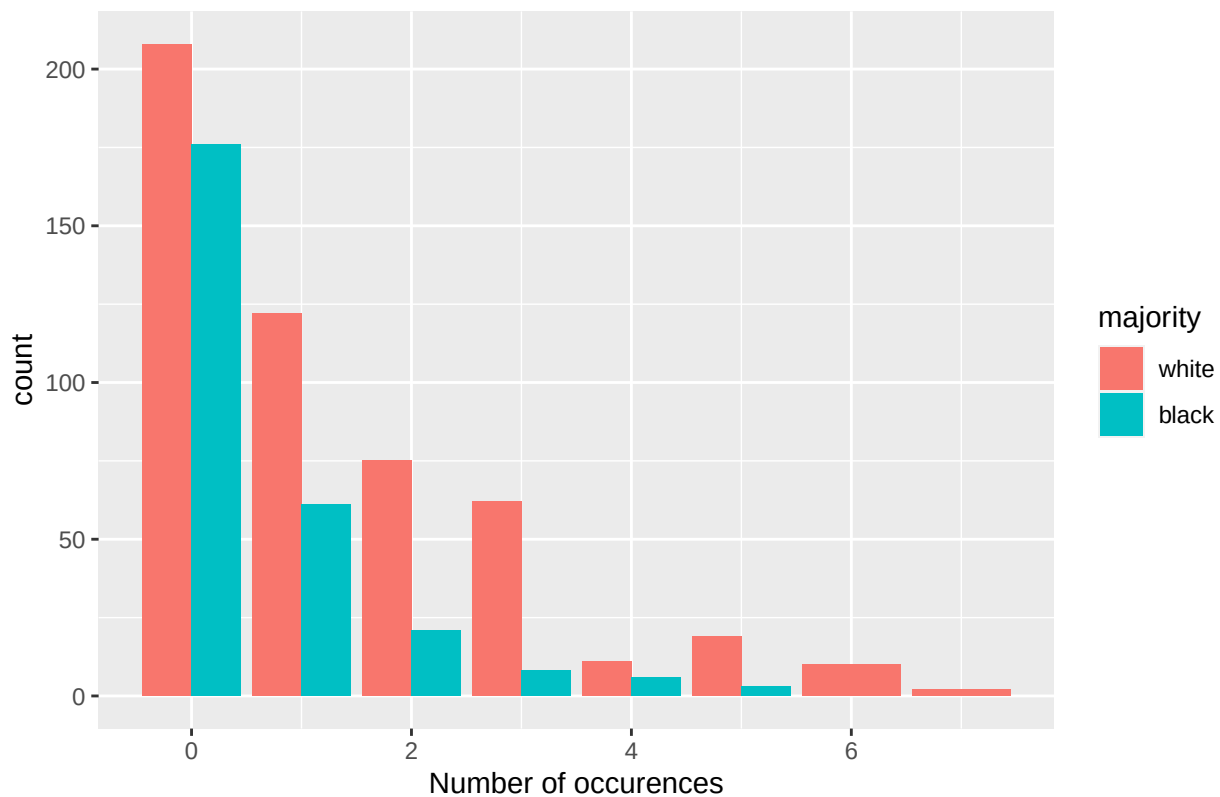
```
##
## Welch Two Sample t-test
##
## data: data[data$majority == "black", ]$shortest_path and data[data$majority == "white", ]$shortest_path
## t = 7.4695, df = 556.68, p-value = 1.566e-13
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 124.8873      Inf
## sample estimates:
## mean of x mean of y
## 834.6204 674.3902
```

Therefore, there is a significant *increase* in the amount of time needed to reach Union Station by individuals coming from Black majority neighborhoods.

Number of cameras en route by demographic group

```
ggplot(data, aes(x = n_cameras, fill = majority)) +
  geom_bar(position = "dodge") +
  labs(x = "Number of occurrences", x = "Number of cameras",
       title = "Number of cameras en route to destination by demographic group")
```

Number of cameras en route to destination by demographic group



```
black_mean_cameras <- mean(data[data$majority == 'black', ]$n_cameras)
white_mean_cameras <- mean(data[data$majority == 'white', ]$n_cameras)

results <- t.test(data[data$majority == 'black', ]$n_cameras, data[data$majority == 'white', ]$n_cameras)

results
```

```
##
## Welch Two Sample t-test
##
## data: data[data$majority == "black", ]$n_cameras and data[data$majority == "white", ]$n_cameras
## t = -7.7616, df = 748.99, p-value = 1
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## -0.8662695 Inf
## sample estimates:
## mean of x mean of y
## 0.6036364 1.3182711
```

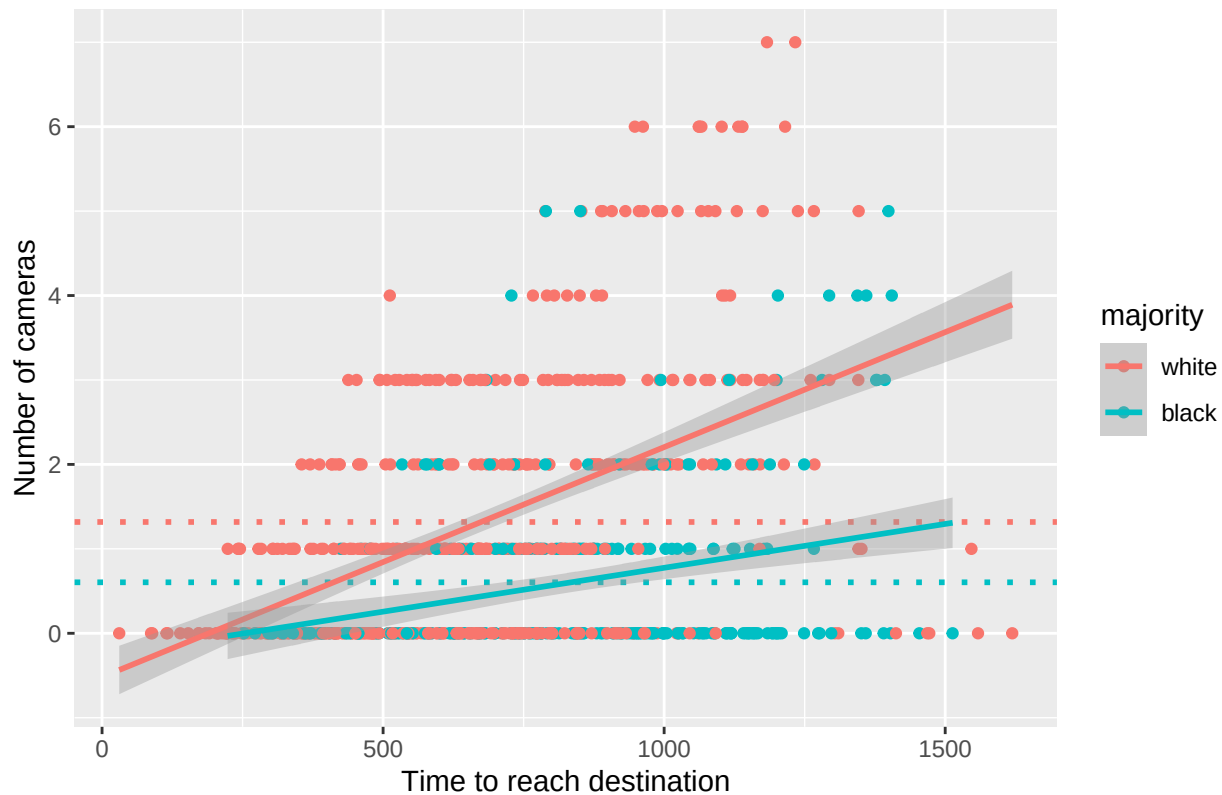
Therefore, there is no significant difference in the number of cameras encountered on the way to Union station by individuals coming from Black majority neighborhoods.

```
ggplot(data, aes(x = shortest_path, y = n_cameras, color = majority)) +
  geom_point() +
  geom_hline(yintercept=white_mean_cameras, linewidth=1, color="#F8766D", linetype="dotted") +
  geom_hline(yintercept=black_mean_cameras, linewidth=1, color="#00bfc4", linetype="dotted") +
  geom_smooth(method="lm") +
  labs(x = "Time to reach destination", y = "Number of cameras",
```

```
title = "Number of cameras en route to destination by demographic group")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Number of cameras en route to destination by demographic group



Dotted line represents group mean.

```
linmod <- lm(data = data, n_cameras ~ majority + shortest_path)
summary(linmod)
```

```
##
## Call:
## lm(formula = n_cameras ~ majority + shortest_path, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.3261 -0.8277 -0.1813  0.5551  4.6010
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.1147235  0.1183579  -0.969   0.333
## majorityblack -1.0551038  0.0961976 -10.968 <2e-16 ***
## shortest_path  0.0021249  0.0001554  13.675 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.242 on 781 degrees of freedom
## Multiple R-squared:  0.2397, Adjusted R-squared:  0.2377
## F-statistic: 123.1 on 2 and 781 DF, p-value: < 2.2e-16
```

```
linmod <- lm(data = data, absolute_increase ~ majority + shortest_path)
summary(linmod)

##
## Call:
## lm(formula = absolute_increase ~ majority + shortest_path, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -86.601 -23.274  -7.604  13.004 194.167
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -8.240672    3.437466  -2.397   0.0167 *
## majorityblack -23.711997    2.793865  -8.487   <2e-16 ***
## shortest_path  0.058569    0.004513  12.979   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 36.06 on 781 degrees of freedom
## Multiple R-squared:  0.2011, Adjusted R-squared:  0.1991
## F-statistic: 98.32 on 2 and 781 DF,  p-value: < 2.2e-16
```

Therefore, there is a significant *decrease* in the number of cameras encountered and additional time when avoiding cameras, when going to Union station by individuals coming from Black majority neighborhoods, when controlling for distance traveled.

O'Hare airport

```
data <- read.csv('./ORD.csv')

data$majority <- relevel(factor(data$majority), ref='white')

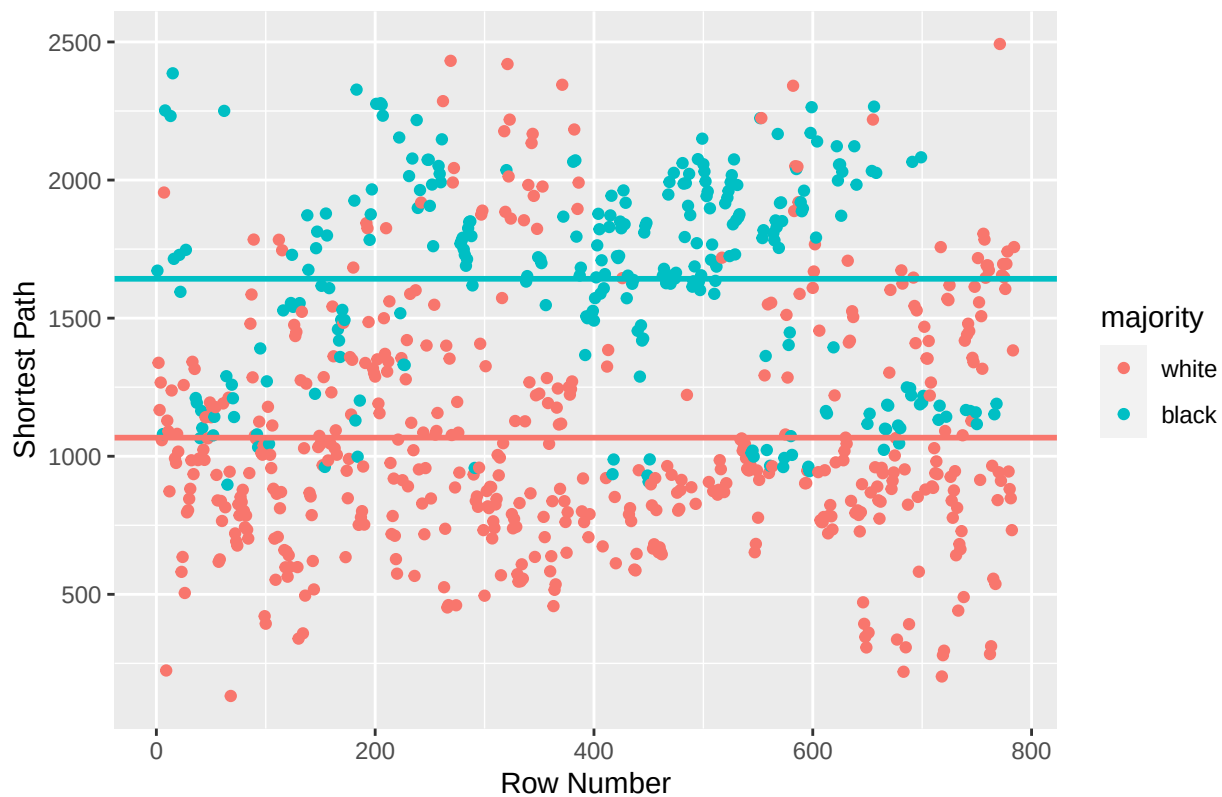
#head(data)

black_mean_ttr <- mean(data[data$majority == 'black', ]$shortest_path)
white_mean_ttr <- mean(data[data$majority == 'white', ]$shortest_path)
```

Mean time to reach by demographic group

```
ggplot(data, aes(x = seq_along(shortest_path), y = shortest_path, color = majority)) +
  geom_point() +
  geom_hline(yintercept=black_mean_ttr, linewidth=1, color="#00bfc4") +
  geom_hline(yintercept=white_mean_ttr, linewidth=1, color="#F8766D") +
  labs(x = "Row Number", y = "Shortest Path",
       title = "Time to reach destination by demographic group")
```

Time to reach destination by demographic group



```
results <- t.test(data[data$majority == 'black', ]$shortest_path, data[data$majority == 'white', ]$shortest_path)
```

```
results
```

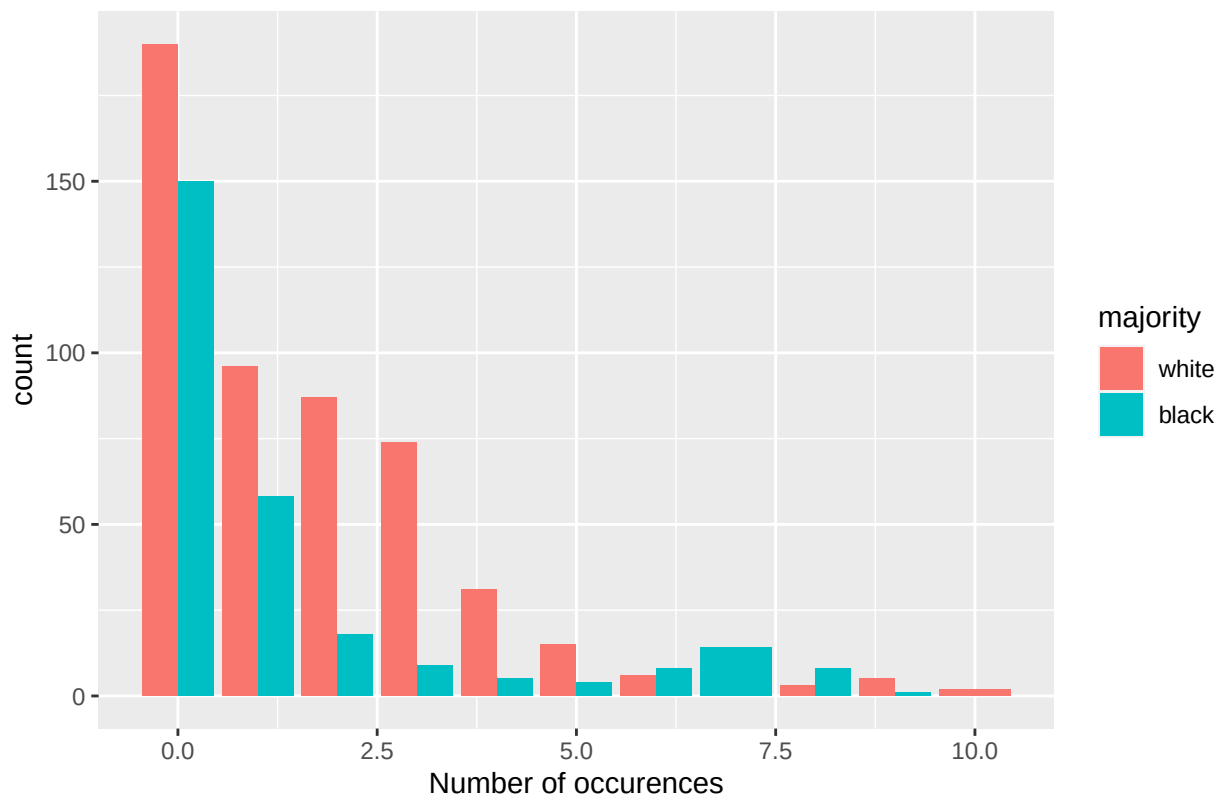
```
##
## Welch Two Sample t-test
##
## data: data[data$majority == "black", ]$shortest_path and data[data$majority == "white", ]$shortest_path
## t = 19.582, df = 640.24, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  526.7678      Inf
## sample estimates:
## mean of x mean of y
## 1642.444 1067.296
```

Therefore, there is a significant *increase* in the amount of time needed to reach O'Hare airport by individuals coming from Black majority neighborhoods.

Number of cameras en route by demographic group

```
ggplot(data, aes(x = n_cameras, fill = majority)) +
  geom_bar(position = "dodge") +
  labs(x = "Number of occurrences", x = "Number of cameras",
       title = "Number of cameras en route to destination by demographic group")
```

Number of cameras en route to destination by demographic group



```
black_mean_cameras <- mean(data[data$majority == 'black', ]$n_cameras)
white_mean_cameras <- mean(data[data$majority == 'white', ]$n_cameras)

results <- t.test(data[data$majority == 'black', ]$n_cameras, data[data$majority == 'white', ]$n_cameras)

results
```

```
##
## Welch Two Sample t-test
##
## data: data[data$majority == "black", ]$n_cameras and data[data$majority == "white", ]$n_cameras
## t = -1.385, df = 464.63, p-value = 0.9166
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## -0.4847003      Inf
## sample estimates:
## mean of x mean of y
## 1.381818 1.603143
```

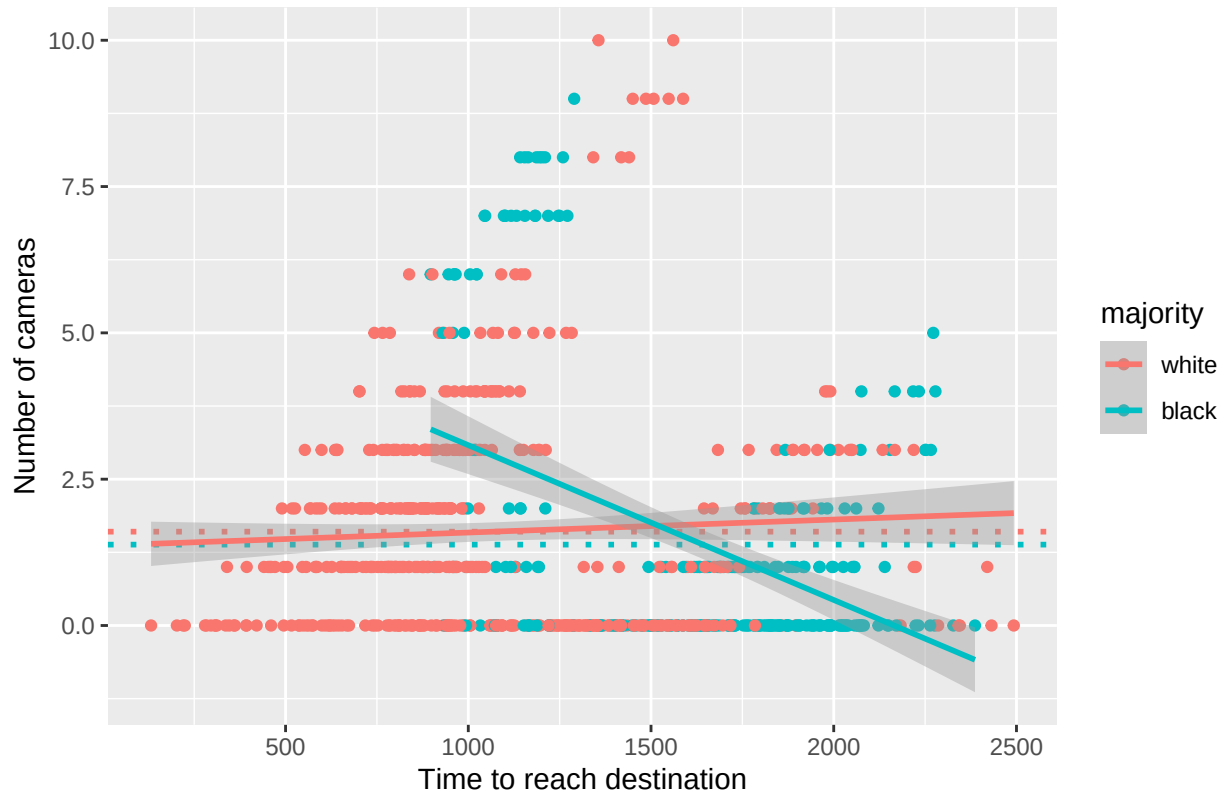
Therefore, there is a no significant difference in the number of cameras encountered on the way to O'Hare airport by individuals coming from Black majority neighborhoods.

```
ggplot(data, aes(x = shortest_path, y = n_cameras, color = majority)) +
  geom_point() +
  geom_hline(yintercept=white_mean_cameras, linewidth=1, color="#F8766D", linetype="dotted") +
  geom_hline(yintercept=black_mean_cameras, linewidth=1, color="#00bfc4", linetype="dotted") +
  geom_smooth(method="lm") +
  labs(x = "Time to reach destination", y = "Number of cameras",
```

```
title = "Number of cameras en route to destination by demographic group")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Number of cameras en route to destination by demographic group



Dotted line represents group mean.

```
linmod <- lm(data = data, n_cameras ~ majority + shortest_path)
summary(linmod)
```

```
##
## Call:
## lm(formula = n_cameras ~ majority + shortest_path, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1533 -1.3967 -0.7265  0.8218  8.6871
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.2309385  0.2040496  10.933  < 2e-16 ***
## majorityblack  0.1169834  0.1785622   0.655  0.512570
## shortest_path -0.0005882  0.0001725  -3.409  0.000684 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.984 on 781 degrees of freedom
## Multiple R-squared:  0.01742,    Adjusted R-squared:  0.0149
## F-statistic: 6.923 on 2 and 781 DF,  p-value: 0.001046
```

```
linmod <- lm(data = data, absolute_increase ~ majority + shortest_path)
summary(linmod)

##
## Call:
## lm(formula = absolute_increase ~ majority + shortest_path, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -51.24  -36.60  -25.27   27.62  194.56
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  52.638131   4.995895  10.536  <2e-16 ***
## majorityblack -3.728051   4.371868  -0.853   0.3941
## shortest_path -0.010571   0.004224  -2.503   0.0125 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 48.57 on 781 degrees of freedom
## Multiple R-squared:  0.01705,    Adjusted R-squared:  0.01453
## F-statistic: 6.773 on 2 and 781 DF,  p-value: 0.001213
```

Therefore, there is a no significant difference in the number of cameras encountered or additional time when avoiding cameras, when going to O'Hare airport by individuals coming from Black majority neighborhoods, when controlling for distance traveled.

Midway airport

```
data <- read.csv('./MDW.csv')

data$majority <- relevel(factor(data$majority), ref='white')

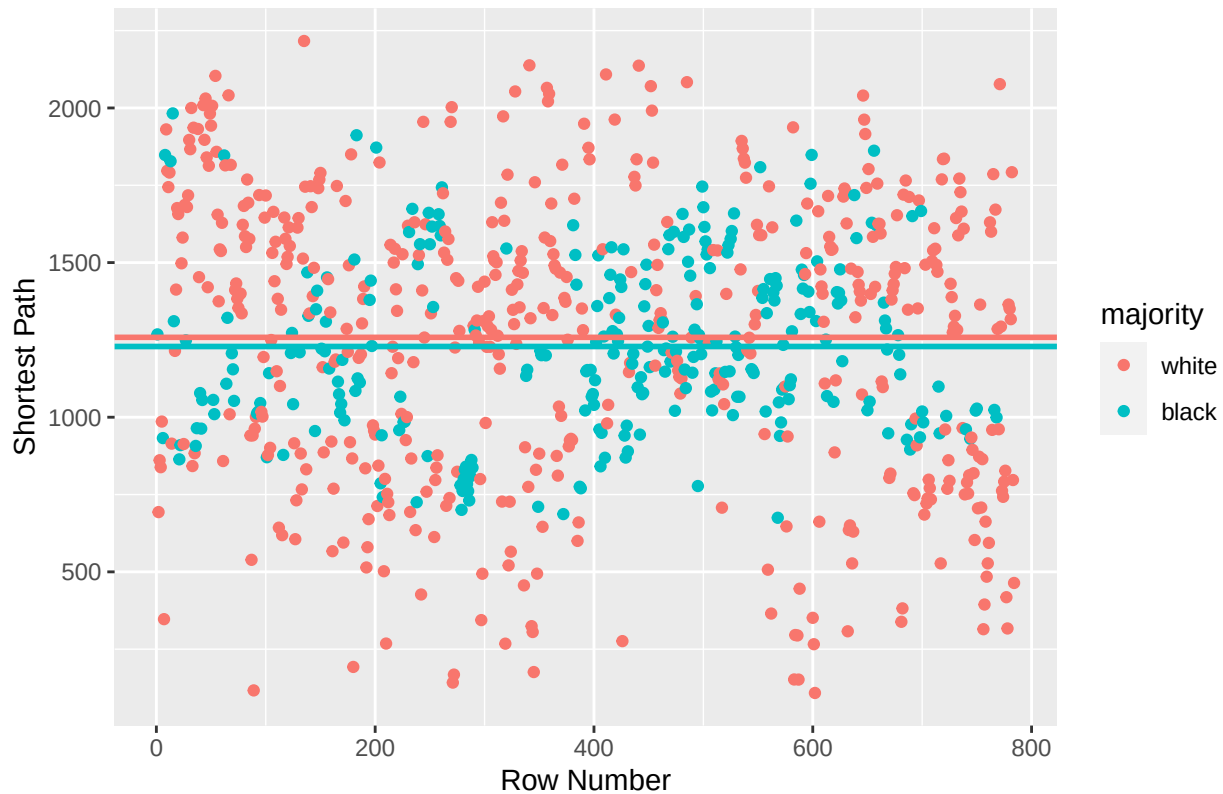
#head(data)

black_mean_ttr <- mean(data[data$majority == 'black', ]$shortest_path)
white_mean_ttr <- mean(data[data$majority == 'white', ]$shortest_path)
```

Mean time to reach by demographic group

```
ggplot(data, aes(x = seq_along(shortest_path), y = shortest_path, color = majority)) +
  geom_point() +
  geom_hline(yintercept=black_mean_ttr, linewidth=1, color="#00bfc4") +
  geom_hline(yintercept=white_mean_ttr, linewidth=1, color="#F8766D") +
  labs(x = "Row Number", y = "Shortest Path",
       title = "Time to reach destination by demographic group")
```

Time to reach destination by demographic group



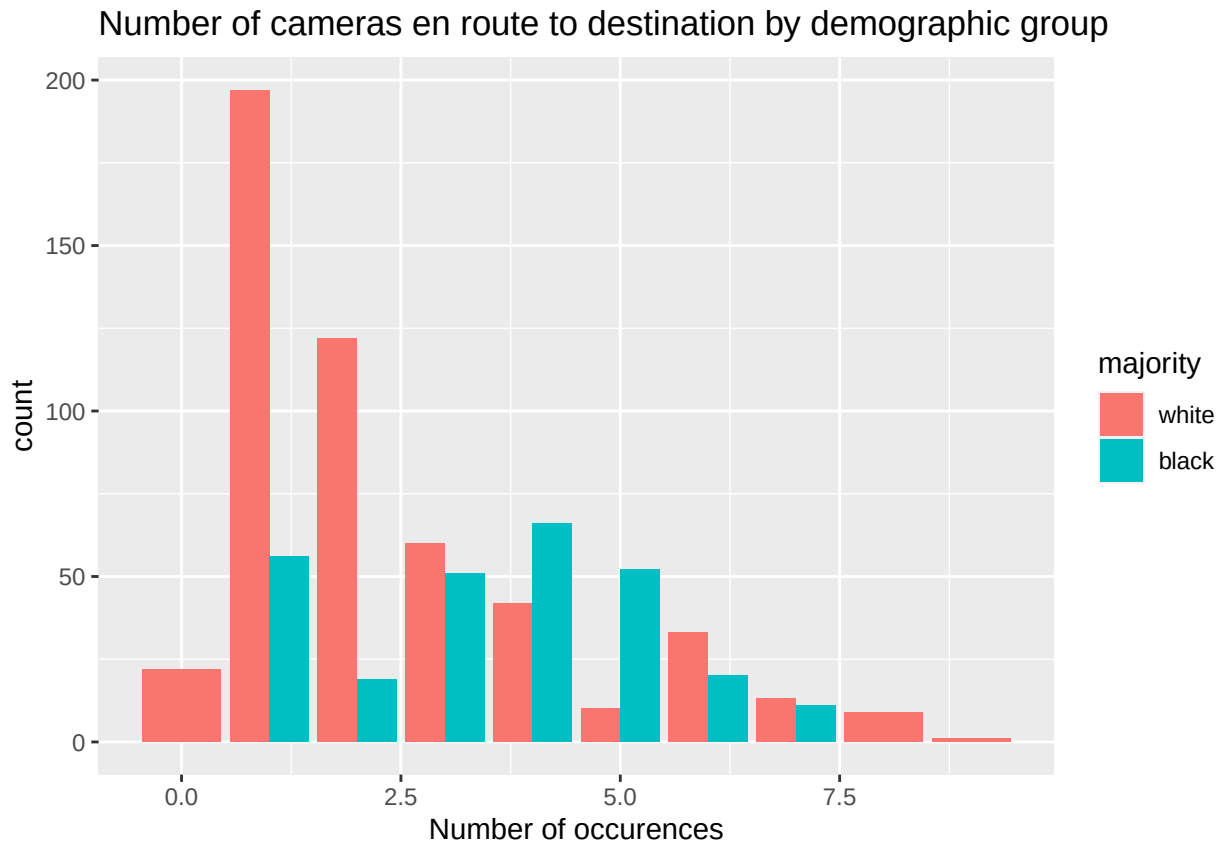
```
results <- t.test(data[data$majority == 'black', ]$shortest_path, data[data$majority == 'white', ]$shortest_path)
results
```

```
##
## Welch Two Sample t-test
##
## data: data[data$majority == "black", ]$shortest_path and data[data$majority == "white", ]$shortest_path
## t = -1.1086, df = 775.53, p-value = 0.866
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## -73.14051      Inf
## sample estimates:
## mean of x mean of y
## 1228.705 1258.132
```

Therefore, there is a no significant difference in the amount of time needed to reach Midway airport by individuals coming from Black majority neighborhoods.

Number of cameras en route by demographic group

```
ggplot(data, aes(x = n_cameras, fill = majority)) +
  geom_bar(position = "dodge") +
  labs(x = "Number of occurrences", x = "Number of cameras",
       title = "Number of cameras en route to destination by demographic group")
```



```
black_mean_cameras <- mean(data[data$majority == 'black', ]$n_cameras)
white_mean_cameras <- mean(data[data$majority == 'white', ]$n_cameras)

results <- t.test(data[data$majority == 'black', ]$n_cameras, data[data$majority == 'white', ]$n_cameras)

results
```

```
##
## Welch Two Sample t-test
##
## data: data[data$majority == "black", ]$n_cameras and data[data$majority == "white", ]$n_cameras
## t = 8.6795, df = 604.08, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 0.9274791 Inf
## sample estimates:
## mean of x mean of y
## 3.520000 2.375246
```

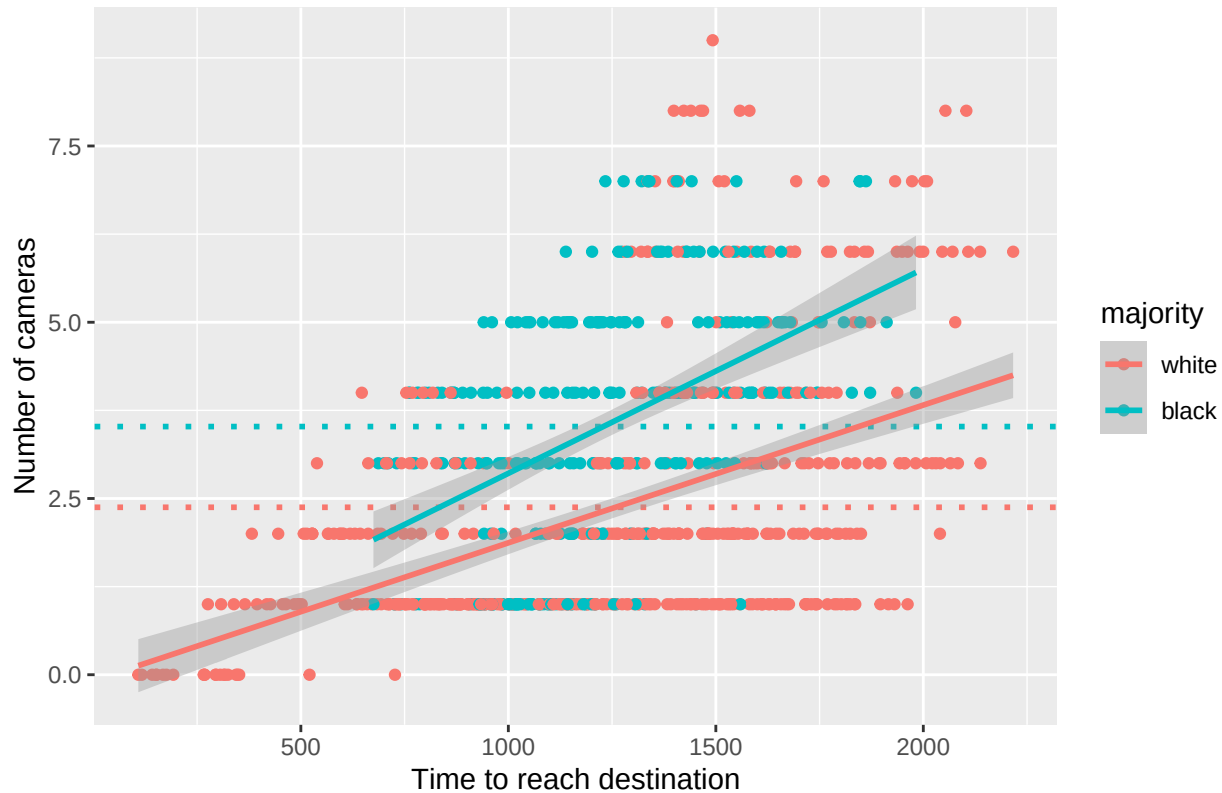
Therefore, there is a significant *increase* in the number of cameras encountered on the way to Midway airport by individuals coming from Black majority neighborhoods, when controlling for distance traveled.

```
ggplot(data, aes(x = shortest_path, y = n_cameras, color = majority)) +
  geom_point() +
  geom_hline(yintercept=white_mean_cameras, linewidth=1, color="#F8766D", linetype="dotted") +
  geom_hline(yintercept=black_mean_cameras, linewidth=1, color="#00bfc4", linetype="dotted") +
  geom_smooth(method="lm") +
  labs(x = "Time to reach destination", y = "Number of cameras",
```

```
title = "Number of cameras en route to destination by demographic group")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Number of cameras en route to destination by demographic group



Dotted line represents group mean.

```
linmod <- lm(data = data, n_cameras ~ majority + shortest_path)
summary(linmod)
```

```
##
## Call:
## lm(formula = n_cameras ~ majority + shortest_path, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.2158 -1.0566 -0.2787  1.0120  6.1315
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.2728415  0.1882369  -1.449   0.148
## majorityblack  1.2066915  0.1190572  10.135 <2e-16 ***
## shortest_path  0.0021048  0.0001387  15.171 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.59 on 781 degrees of freedom
## Multiple R-squared:  0.2924, Adjusted R-squared:  0.2906
## F-statistic: 161.4 on 2 and 781 DF, p-value: < 2.2e-16
```

```
linmod <- lm(data = data, absolute_increase ~ majority + shortest_path)
summary(linmod)
```

```
##
## Call:
## lm(formula = absolute_increase ~ majority + shortest_path, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -64.13  -28.46  -10.54   17.34  181.20
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.288019   4.896541  -0.263    0.793
## majorityblack 19.633340   3.096992   6.339 3.9e-10 ***
## shortest_path  0.036992   0.003609  10.250 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41.36 on 781 degrees of freedom
## Multiple R-squared:  0.1529, Adjusted R-squared:  0.1507
## F-statistic: 70.48 on 2 and 781 DF,  p-value: < 2.2e-16
```

Therefore, there is a significant *increase* in the number of cameras encountered and additional time when avoiding cameras, when going to Midway airport by individuals coming from Black majority neighborhoods, when controlling for distance traveled.